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Solutions to

GRAPHICAL SUMMARIES WITH MCQS

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SUBJECTS ONLINE

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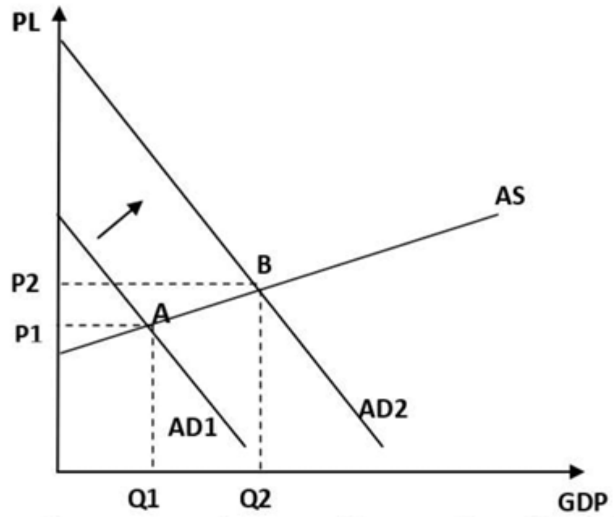
Public Finance II — Graphical Summaries with MCQs

📊 GRAPH 1: Government Consumption Expenditure with Elastic Supply (Figure 2)

The Graph:



Increase the government consumption spending with elastic supply
(Figure2)

Key Information:

- Horizontal axis: GDP (Gross Domestic Product / National Income)
- Vertical axis: PL (Price Level of national production)
- AD: Aggregate Demand = C + I + G (closed economy)
- AS: Aggregate elastic supply — shows positive relationship between price level and real GDP in short run
- Before: equilibrium at point A → income = Q1, price = P1
- After increasing government consumption expenditure: AD shifts upward from AD1 to AD2
- New equilibrium at point B → income = Q2, price = P2
- **Noticeable: increase in national income (Q1→Q2) is LARGER than increase in price (P1→P2)**

شرح الرسمة : الرسمة دي بتوضح اللي بيحصل لما الحكومة تزود إنفاقها الاستهلاكي في اقتصاد عنده عرض مرن (زي البلد المتقدمة اللي فيها طاقة إنتاجية فاضية).
المحور الأفقي: الناتج القومي (GDP). المحور الرأسي: مستوى الأسعار. منحنى AS مرن يعني الميله خفيف — الاقتصاد يقدر يزود الإنتاج من غير ما يرفع أسعاره كتير.
لما الحكومة تزود الإنفاق ← AD بتتحرك لفوق ← نقطة التوازن تتحول من A لـ B ← الدخل بيزيد من Q1 لـ Q2 والأسعار بترتفع من P1 لـ P2. المهم هنا إن الزيادة في الدخل أكبر من الزيادة في الأسعار. ده معناه إن الإنفاق الحكومي هنا شغال كـ expansionary fiscal policy ومفيد جداً وقت الركود.

MCQs:

Q1. In Figure 2, the initial equilibrium before increasing government consumption expenditure is at:

A) Point B where AD2 intersects AS

B) Point A where AD1 intersects the elastic AS curve at price P1 and income Q1

C) The origin where GDP equals zero

D) Point P2 on the price level axis
Answer: B

Q2. After increasing government consumption expenditure with elastic supply, the Aggregate Demand curve:

A) Shifts downward from AD2 to AD1

B) Becomes perfectly inelastic

C) Shifts upward from AD1 to AD2 by the amount of government spending

D) Remains unchanged at AD1
Answer: C

Q3. In the elastic supply case (Figure 2), which of the following is noticeable?

A) The increase in price is larger than the increase in national income

B) National income decreases while prices increase

C) The increase in national income is larger than the increase in price

D) Both national income and price remain unchanged
Answer: C

Q4. The Aggregate Supply (AS) curve in Figure 2 shows:

A) A negative relationship between price level and real GDP

B) A positive relationship between price level and real GDP in the short run

C) A perfectly horizontal line indicating zero inflation

D) A perfectly vertical line indicating full employment
Answer: B

Q5. Government consumption spending with an elastic supply curve is most effective as:

A) A contractionary fiscal policy tool during inflation

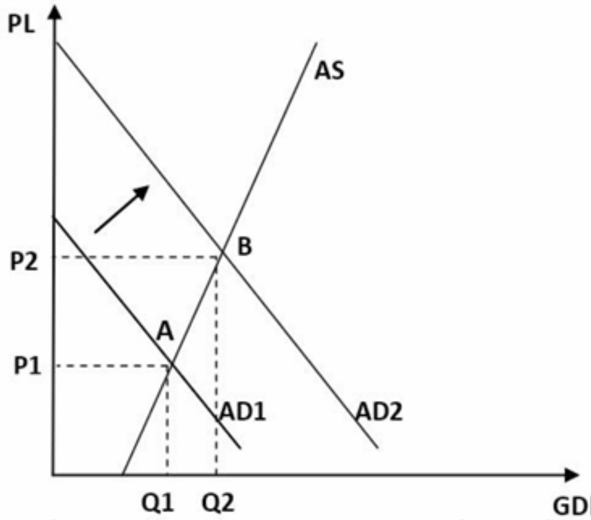
B) An expansionary fiscal policy to boost economic growth during recession

C) A tool to reduce aggregate demand

D) A method to shift the AS curve downward
Answer: B

📊 GRAPH 2: Government Consumption Expenditure with Inelastic Supply (Figure 3)

The Graph:



Increase the government consumption spending with inelastic supply
(Figure3)

Key Information:

- AS curve is steep (inelastic) — economy is close to full capacity, as in developing countries
- Before: equilibrium at point A → income = Q1, price = P1
- After increasing government consumption expenditure: AD shifts upward from AD1 to AD2
- New equilibrium at point B → income = Q2, price = P2
- **Noticeable: increase in PRICE (P1→P2) is LARGER than increase in national income (Q1→Q2)**
- Higher government spending causes inflationary pressures and only a little increase in real GDP
- **Key conclusion: "The effect of government consumption spending depends on how responsive supply is"**

شرح الرسمة : الرسمة دي نفس الفكرة بس في اقتصاد عنده عرض غير مرن (steep AS) — وده بيحصل في البلد النامية اللي بتشتغل قريب من طاقتها القصوى.
لما الحكومة تزود الإنفاق → AD بتتحرك لفوق بنفس المقدار زي قبل → لكن لأن AS شبه رأسية، الاقتصاد مش قادر يزود الإنتاج بسهولة → فبدل ما الزيادة تيجي في الكميات (GDP)، بتيجي في الأسعار. النتيجة: أسعار بترتفع كتير وزيادة في الدخل الحقيقي قليلة.
خلاصة الرسمتين مع بعض: أثر الإنفاق الحكومي بيتوقف على مرونة العرض — مرن → دخل يزيد أكتر. غير مرن → أسعار ترتفع أكتر.

MCQs:

Q1. In Figure 3, the AS curve is steep (inelastic) because:

A) The economy has large amounts of unused capacity

B) The economy is close to full capacity, as in developing countries

C) The government has imposed price controls

D) Aggregate demand is falling
Answer: B

Q2. With an inelastic supply curve, increasing government consumption expenditure mainly results in:

A) A large increase in real GDP with stable prices

B) A decrease in aggregate demand

C) Inflationary pressures and only a little increase in real GDP

D) A downward shift of the AS curve
Answer: C

Q3. Comparing Figure 2 (elastic) and Figure 3 (inelastic), the key difference in outcome is:

A) In elastic supply, price rises more; in inelastic supply, income rises more

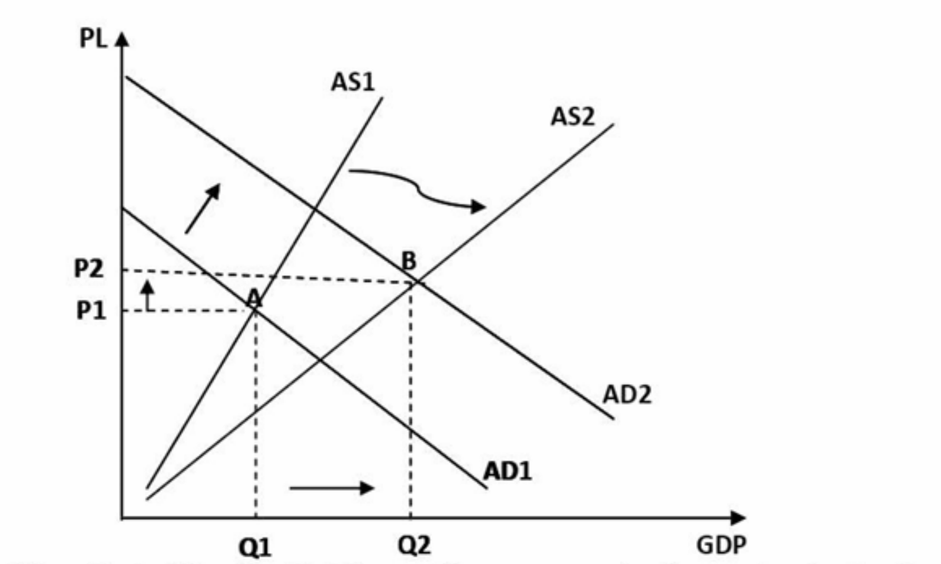
B) In elastic supply, income rises more than price; in inelastic supply, price rises more than income

C) Both cases produce identical results in terms of income and price changes

D) In inelastic supply, both income and

📊 GRAPH 3: Effect of Capital (Investment) Expenditure on Production and Price Level (Figure 4)

The Graph:



The effect of the Capital Expenditures on production and price level
(Figure4)

Key Information:

- Two AS curves: AS1 (before) and AS2 (after capital expenditure takes effect)
- Before: equilibrium at point A → AD1 cuts AS1 → income = Q1, price = P1
- After capital expenditure: AD shifts up from AD1 to AD2 (immediate effect)
- After a period of time: AS shifts downward/rightward from AS1 to AS2 (because infrastructure and human capital improve productivity)
- New equilibrium at point B → AD2 cuts AS2 → income = Q2, price = P2
- **Key distinction from consumption expenditure: Capital expenditure affects BOTH AD and AS**
- The economy can produce more domestic product at higher quality and lower cost

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شرح الرسمة : الرسمة دي بتوضح الفرق المهم بين الإنفاق الاستثماري والإنفاق الاستهلاكي.
الإنفاق الاستثماري (زي بناء طرق ومستشفيات وتعليم) بيعمل حاجتين مش حاجة واحدة:
الأول:

زي الإنفاق الاستهلاكي بالظبط ← بيرفع AD من AD1 لـ AD2 فوراً.
الثاني — والمهم:

بعد فترة، التحسين في البنية التحتية والرأس المال البشري بيخلي الاقتصاد يشتغل بكفاءة أعلى ← AS بيتحرك من AS1 لـ AS2 (ينزل لأسفل/يمين) ← يعني الاقتصاد بيقدر ينتج أكتر بتكلفة أقل وجودة أحسن.
النتيجة النهائية:

دخل قومي أعلى (Q2) وأسعار معتدلة (P2) — أحسن من الإنفاق الاستهلاكي البحت.

MCQs:
Q1. Unlike government consumption expenditure, investment (capital) expenditure affects:
A- Only the Aggregate Demand curve
B- Only the price level
C- Both Aggregate Demand and Aggregate Supply curves
D- Only employment levels without affecting GDP
Answer: C

Q2. In Figure 4, after a period of time following capital expenditure, the AS curve:
A- Shifts upward from AS2 to AS1
B- Remains fixed at AS1
C- Shifts downward/rightward from AS1 to AS2 because productivity improves
D- Becomes perfectly inelastic
Answer: C

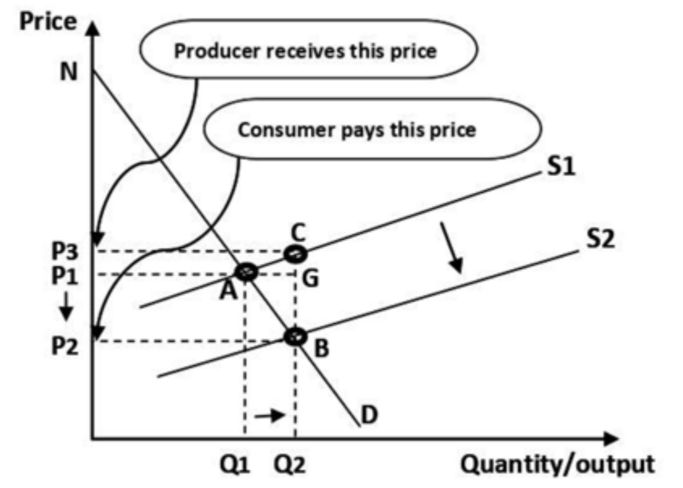
Q3. The reason AS becomes more elastic after capital expenditure is:
A- The government reduces taxes on producers
B- Positive effects of infrastructure and human capital investment improve productivity and lower costs
C- Aggregate demand decreases over time
D- The exchange rate appreciates
Answer: B

Q4. Examples of government capital expenditure that improve human capital include:
A- Paying civil servants' salaries
B- Spending on defense and police
C- Spending on health, education, training, and research
D- Transfer payments to unemployed individuals
Answer: C

Q5. The final equilibrium in Figure 4 (point B) is reached where:
A- AD1 intersects AS1
B- AD2 intersects AS2
C- AD1 intersects AS2
D- The price level falls to zero
Answer: B

📊 GRAPH 4: Economic Effects of Production Subsidy — Case 1: Elastic (Increasing) Supply

The Graph:



"The economic effects of production subsidy"

Key Information:

- Before subsidy: equilibrium at A (price P1, quantity Q1); supply curve S1
- Consumer surplus before = ΔNAP1
- Government gives unit subsidy → marginal cost decreases → S1 shifts DOWN to S2 (parallel shift by subsidy amount)
- New equilibrium at B (price P2, quantity Q2); supply curve S2
- Subsidy per unit (BC) = vertical distance between S1 and S2
- Consumer price falls from P1 to P2 (consumer pays P2)
- Producer receives P3 = P2 + subsidy per unit BC
- Total subsidy cost = CBP2P3 = Q2 × BC

- Consumer surplus after = ΔNBP2
- Change in consumer surplus (welfare gain) = trapezoid ABP1P2
- **Benefit split:**
  - Benefit to consumer = GB = P1P2 per unit → total area P1GBP2
  - Benefit to producer = CG = P3P1 per unit → total area CGP1P3
  - **Producer benefit < Consumer benefit** (because supply is elastic)

شرح الرسمة : الرسمة دي بتوضح أثر الدعم الإنتاجي (unit subsidy) لما العرض يكون مرن (منحنى مائل).
قبل الدعم: التوازن عند A (سعر P1، كمية Q1).
الحكومة بتدي المنتج دعم لكل وحدة → التكلفة الحدية بتنخفض → S1 بتنزل لـ S2.
بعد الدعم: التوازن الجديد عند B (سعر أقل P2، كمية أكبر Q2).
المستهلك بيدفع P2 (أقل من P1) والمنتج بيستلم P3 (= P2 + الدعم). الفرق بين P3 وP1 هو نصيب المنتج من الدعم، والفرق بين P1 وP2 هو نصيب المستهلك.
النتيجة المهمة: لأن العرض مرن، المستهلك بياخد جزء أكبر من فائدة الدعم من المنتج. التكلفة الكلية للدعم على الحكومة = Q2 × مقدار الدعم لكل وحدة.

MCQs:

Q1. When the government gives a unit subsidy to the producer, the supply curve shifts:
A- Upward from S2 to S1 by the amount of the subsidy
B- Downward in parallel from S1 to S2 by the amount of the subsidy
C- Rightward only without changing the price intercept
D- Remains at S1 with only quantity changing
Answer: B

Q2. In the subsidy diagram with elastic supply, the subsidy per unit equals:
A- The horizontal distance between S1 and S2
B- The distance between P1 and P2 only
C- The vertical distance BC between the two supply curves
D- The area of triangle NAP1
Answer: C

Q3. The total (overall) cost of the government subsidy is calculated as:

A- Q1 × subsidy per unit BC

B- Q2 × subsidy per unit BC (area CBP2P3)

C- The change in consumer surplus only

D- P3 – P1 × Q1

Answer: B

Q4. With elastic supply, the benefit of the subsidy is distributed such that:

A- The producer gets all the benefit

B- The consumer gets all the benefit

C- The benefit is shared, with the consumer getting more than the producer

D- The benefit is shared equally between consumer and producer

Answer: C

Q5. In the elastic supply subsidy diagram, the consumer price after subsidy is:

A- P3 (the producer price)

B- P1 (the original equilibrium price)

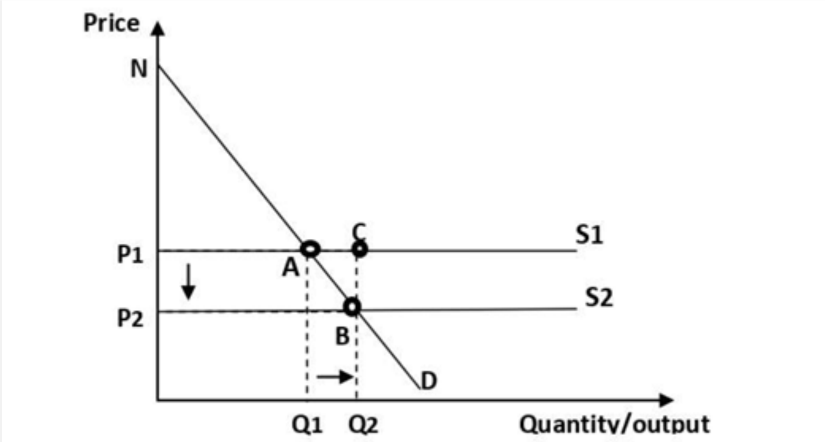
C- P2 (lower than P1 by the consumer's share of subsidy)

D- Zero

Answer: C

📊 GRAPH 5: Production Subsidy — Case 2: Perfectly Elastic (Constant Cost) Supply

The Graph:



Key Information:

- Supply curve S1 is perfectly horizontal (perfectly elastic = constant marginal cost)
- Before subsidy: equilibrium at A (price P1, quantity Q1)
- After unit subsidy: S shifts down to S2 (also horizontal, lower by full subsidy amount BC)
- New equilibrium at B (price P2, quantity Q2)
- Consumer price falls from P1 to P2 by the FULL amount of the subsidy (BC = P1P2)
- Producer price remains at P1 — producer receives NO benefit from subsidy
- **Entire subsidy benefit goes to consumer**
- Total subsidy cost = BCP1P2 = Q2 × BC
- **Key Rule: The more elastic the supply, the less benefit of subsidy to producer**
- In perfectly elastic case → producer benefit = ZERO

شرح الرسمة : الحالة الثانية:

لما العرض يكون مرن تماماً (horizontal) — يعني المنتج بيشتغل بتكلفة ثابتة.
لما الحكومة تدي دعم → S بتنزل بالكامل بمقدار الدعم. السعر بيقل من P1 لـ P2 بكامل مقدار الدعم.
المستهلك بياخد كل الفايدة. المنتج مش بياخد أي فايدة خالص.
ليه؟ لأن منحنى العرض أفقي تماماً — المنتج مستعد يبيع بأي سعر السوق يحدده، فالدعم كله اتحوّل للمستهلك في صورة سعر أقل.
الخلاصة العامة من Case 1 و Case 2: **كل ما العرض أكثر مرونة، أقل فائدة للمنتج وأكثر فائدة للمستهلك.**

MCQs:

Q1. In Case 2 (perfectly elastic supply), after the unit subsidy is given:
A- The consumer price falls by only part of the subsidy amount
B- The consumer price falls by the full amount of the subsidy
C- The producer receives all the benefit
D- Both consumer and producer share the subsidy equally
Answer: B

Q2. With a perfectly elastic supply curve, the benefit of the subsidy to the producer is:
A- Equal to the full subsidy per unit
B- Greater than the consumer's benefit
C- Zero — the producer receives no benefit
D- Half the subsidy per unit
Answer: C

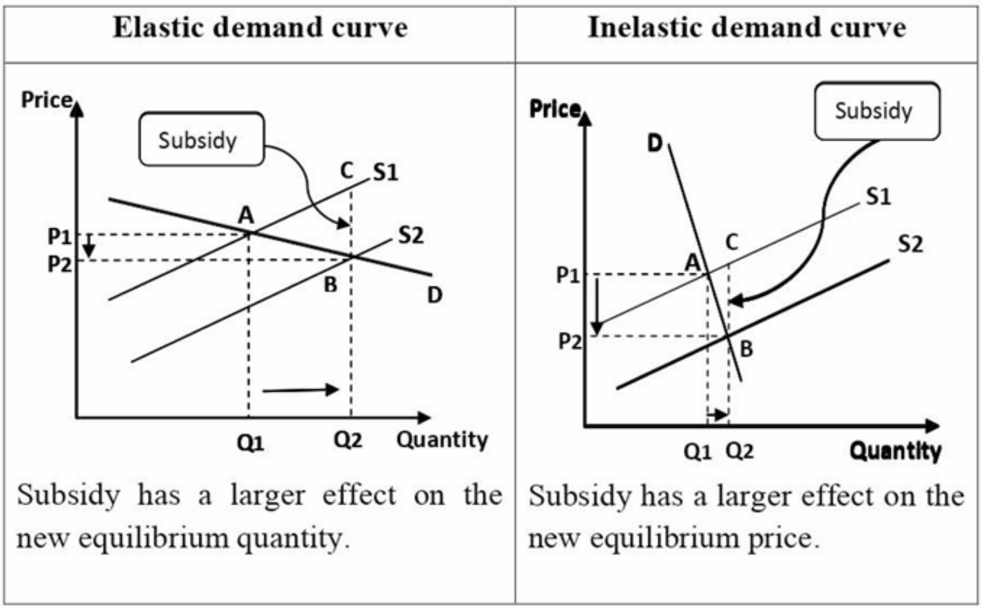
Q3. The key rule derived from comparing Case 1 and Case 2 is:
A- The more inelastic the supply, the less benefit to the producer
B- The more elastic the supply, the less benefit of subsidy to the producer
C- Supply elasticity has no effect on subsidy benefit distribution
D- Perfectly elastic supply gives producers maximum benefit
Answer: B

Q4. In the perfectly elastic supply case, the supply curve S2 after subsidy is:
A- Steeper than S1
B- A perfectly horizontal line lower than S1 by the full subsidy amount
C- The same as S1 but shifted rightward
D- An upward-sloping curve
Answer: B

Q5. The total cost of the government subsidy in Case 2 equals:
A- Q1 × BC
B- Q2 × BC (area BCP1P2)
C- The area of triangle NAP1
D- P1 × Q2 only
Answer: B

📊 GRAPH 6: Effect of Subsidy as Demand Elasticity Changes

The Graph:



Key Information:

- Same subsidy applied in both cases — only demand elasticity differs
- **Elastic demand:** subsidy mainly increases quantity (Q1→Q2 is large); price reduction (P1→P2) is small
- **Inelastic demand:** subsidy mainly reduces price (P1→P2 is large); quantity increase (Q1→Q2) is small
- When demand is perfectly inelastic → consumer gains ALL benefit through lower price
- **Key Rule: The more elastic the demand, the less benefit of subsidy to consumer**

شرح الرسمة : الرسمة دي بتجاوب على سؤال: إيه أثر مرونة الطلب على توزيع فائدة الدعم؟
لو الطلب مرن (الجانب الأيسر): الناس حساسة للسعر ← لما السعر بينخفض، الكمية المطلوبة بتزيد كتير. فالدعم بيأثر أساساً على الكمية.
لو الطلب غير مرن (الجانب الأيمن): الناس مش حساسة للسعر ← لما السعر بينخفض، الكمية ما بتزيدش كتير. فالدعم بيأثر أساساً على السعر — يعني المستهلك بياخد فائدة أكبر في صورة تخفيض سعر.
الخلاصة: **كل ما الطلب أقل مرونة (inelastic)، أكتر فائدة للمستهلك من الدعم. وكل ما الطلب أكتر مرونة، أقل فائدة للمستهلك.**

MCQs:

Q1. When demand is elastic, the main effect of a production subsidy is:
A- A large reduction in equilibrium price with little change in quantity
B- A large increase in equilibrium quantity with a relatively small price reduction
C- No change in either price or quantity
D- A decrease in consumer surplus
Answer: B

Q2. When demand is inelastic, the main effect of a production subsidy is:
A- A large increase in equilibrium quantity
B- No change in price
C- A larger reduction in equilibrium price with a small increase in quantity
D- The producer receives all the benefit
Answer: C

Q3. The key rule about demand elasticity and subsidy benefit states:
A- The more elastic the demand, the more benefit to the consumer
B- The more elastic the demand, the less benefit of subsidy to the consumer
C- Demand elasticity has no effect on how subsidy benefits are distributed
D- Inelastic demand means the producer gains all the benefit
Answer: B

Q4. When demand is perfectly inelastic, the consumer gains:

A- Zero benefit from the subsidy

B- Only half the subsidy benefit

C- All the benefit from the subsidy through a lower price

D- The same benefit as the producer

Answer: C

Q5. Comparing the two panels in Graph 6, the subsidy has a larger effect on the new equilibrium PRICE when:

A- Demand is elastic

B- Supply is perfectly elastic

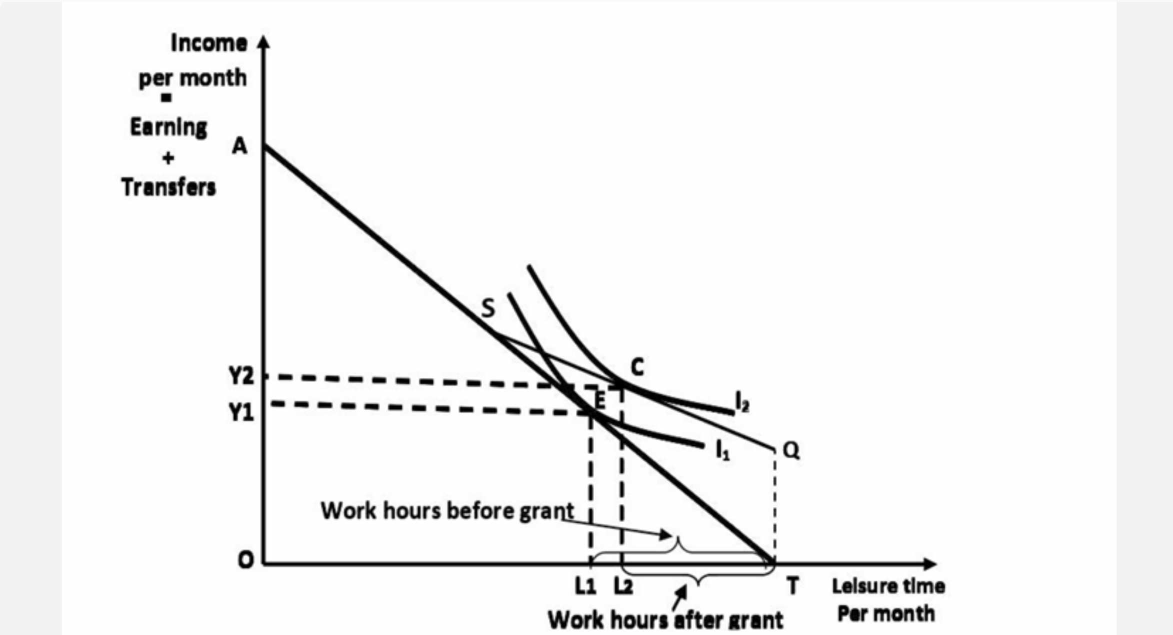
C- Demand is inelastic

D- Both demand and supply are elastic

Answer: C

📊 GRAPH 7: Labor Supply Decision Under Unemployment Grant

The Graph:



Key Information:

created with craft

- Horizontal axis: Leisure time per month (OT); any point also indicates hours of work
- Vertical axis: Income per month (Earning + Transfers)
- Point T: zero working hours AND zero income (before grant)
- Point Q: zero working hours, receives basic grant G = $300
- Grant reduction rate T = 25% → budget line is kinked (ASQ shape)
- The budget line goes from Q upward along a flatter slope until point S, then continues along original slope to A
- At point S: earnings reach breakeven level (G/T = 300/0.25 = $1200) → benefit becomes zero
- I_1, I_2: indifference curves representing preference for leisure vs. income
- **Before grant:** utility maximized at E → working hours = L1T, leisure = OL1, income = Y1
- **After grant:** utility maximized at C → working hours = L2T (LESS than L1T), leisure = OL2, income = Y2 (HIGHER than Y1)
- **Key conclusion: The government welfare program REDUCES the work incentive**

شرح الرسمة : الرسمة دي بتوضح أثر إعانة البطالة على قرار الشخص: يشتغل كام ساعة؟

المحور الأفقي: وقت الفراغ (leisure) — كل ما اتجهنا لليسار كل ما ساعات العمل زادت. المحور الرأسي: الدخل الشهري (كسب + تحويلات).

قبل الإعانة: الشخص بيختار النقطة E على منحنى السواء I_1 → بيشتغل L1T ساعة، دخله Y1.

بعد الإعانة: خط الميزانية اتكسر (kinked) وبقى ASQ:

- عند Q (صفر عمل): بياخد 300 دولار إعانة.
- كل ما بيكسب، الإعانة بتقل بنسبة 25%.
- عند S: الإعانة بتوصل لصفر.

الشخص دلوقتي بيختار النقطة C على منحنى سواء أعلى I_2 → بيشتغل L2T ساعة فقط (أقل من قبل)، لكن دخله Y2 (أعلى من قبل).
النتيجة: الإعانة خلّت الشخص يشتغل أقل لأن دخله الإجمالي بقى مقبول وهو شغال ساعات أقل. **برامج الرفاه ممكن تقلل الحافز على العمل.**

MCQs:

Q1. In the unemployment grant graph, point Q on the horizontal axis represents:
A- Maximum working hours with maximum income
B- Zero working hours where the person receives only the basic grant G
C- The breakeven point where benefit becomes zero
D- The point of maximum utility before the grant
Answer: B

Q2. The budget line becomes kinked (ASQ shape) after the unemployment grant because:
A- The wage rate increases after receiving the grant
B- The grant reduces the effective wage rate earned during the benefit phase, creating a flatter slope until point S
C- The government imposes a tax on all earnings
D- Leisure becomes more expensive after the grant
Answer: B

Q3. Point S on the kinked budget line represents:
A- The point where the person stops working entirely
B- The breakeven earnings level where benefit received equals zero (E = G/T)
C- The maximum income achievable under the grant program
D- The initial equilibrium before the grant
Answer: B

Q4. After the introduction of the unemployment grant, the person moves from point E to point C, which means:
A- Working hours increased from L2T to L1T
B- Income decreased from Y2 to Y1
C- Working hours decreased (L2T < L1T) while total income increased (Y2 > Y1)
D- The person reached a lower indifference curve I_1
Answer: C

Q5. The main conclusion of the unemployment grant labor supply diagram is:

A- Welfare programs always increase the number of hours worked

B- The government welfare program could reduce the work incentive

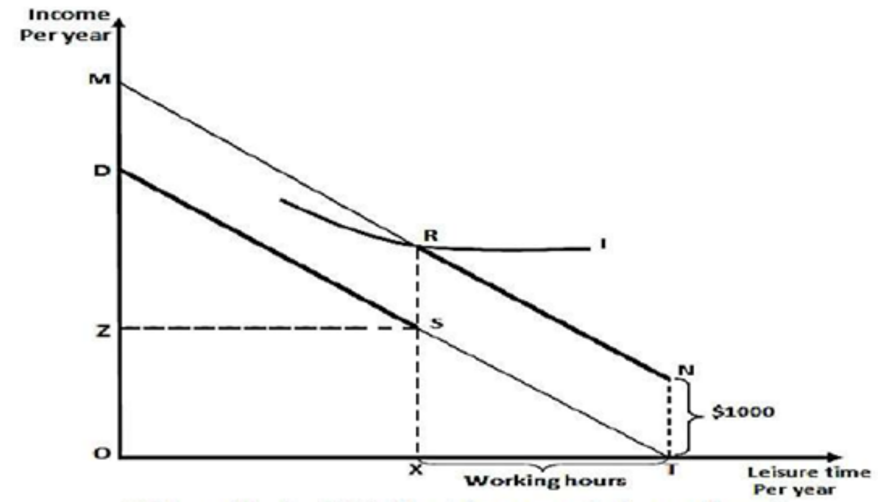
C- Grant programs have no effect on labor supply decisions

D- Higher grants always lead to more employment

Answer: B

📊 GRAPH 8: Effect of Medicaid on Work Incentives

The Graph:



"The effect of Medicaid on work incentives."

Key Information:

- Horizontal axis: Leisure time per year (OT); also represents working hours
- Vertical axis: Income per year
- DT: original budget line before Medicaid
- Medicaid value = $1,000 per year
- At zero working hours (point T) → person receives in-kind income $1,000 → represented by point N
- Moving left from N (working more): income rises by wage rate, same slope as DT → segment NR
- At XT hours of work: earnings reach Z dollars → person LOSES Medicaid → drops from R to S (loses $1,000 instantly)
- Moving further left from S: earns only wage → segment SD

- Budget line with Medicaid = **NRSD** (kinked with a notch at point R)
- **The notch at R creates a strong disincentive:** earning one dollar more than Z causes loss of $1,000
- **Key conclusion: Medicaid creates work disincentives to keep earnings below the cut-off level Z**

شرح الرسمة : الرسمة بتوضح ازاي الميديكيد (التأمين الصحي للفقراء) بيأثر على قرار الشخص إنه يشتغل.
قبل الميديكيد: خط الميزانية العادي DT.
بعد الميديكيد: عند صفر عمل (نقطة T)، الشخص بياخد قيمة الميديكيد 1000 دولار في السنة → نقطة N.
كل ما بيشتغل أكتر، دخله بيزيد بمعدل الأجر → خط NR.
لما يوصل لـ XT ساعة عمل ودخله يوصل Z دولار → بيخسر الميديكيد فوراً → دخله بينزل من R لـ S (يخسر 1000 دولار).
بعد كده من S لـ D بيشتغل عادي بدون ميديكيد.
خط الميزانية الكلي = NRSD فيه "notch" (كسرة حادة) عند R.
النتيجة: الشخص بيفضل يفضل دخله تحت Z عشان ميخسرش الميديكيد — يعني الميديكيد بيخليه يشتغل أقل. ده **work disincentive.**

MCQs:

Q1. In the Medicaid work incentives graph, point N represents:
A- The point of maximum income from working full time
B- Zero working hours where the person receives in-kind Medicaid income of $1,000
C- The breakeven point where Medicaid eligibility ends
D- The original equilibrium before Medicaid was introduced
Answer: B

Q2. The budget line in the presence of Medicaid takes the shape NRSD because:
A- The government taxes all income above Z dollars
B- At earnings level Z, the person immediately loses $1,000 Medicaid benefit, creating a notch (drop from R to S)
C- The wage rate decreases after receiving Medicaid
D- The person voluntarily reduces working hours at point R
Answer: B

Q3. The "notch" at point R in the Medicaid diagram is economically significant because:

A- It represents the highest achievable income under Medicaid

B- Earning one dollar more than Z causes an immediate loss of $1,000, making it irrational to earn just above Z

C- It shows the point where the person maximizes leisure time

D- It represents the government's maximum subsidy payment

Answer: B

Q4. The conclusion "Medicaid creates work disincentives to keep earnings below the cut-off level Z" means:

A- Medicaid encourages people to work more hours to maintain eligibility

B- People rationally choose to keep their earnings below Z to avoid losing $1,000 in Medicaid benefits

C- The government intentionally discourages work through Medicaid

D- Only high-income earners are affected by Medicaid disincentives

Answer: B

Q5. The segment NR of the budget line NRSD has the same slope as the original budget line DT because:

A- The government imposes a tax on income in this range

B- The wage rate is the same — the person earns at the normal wage rate while still receiving Medicaid

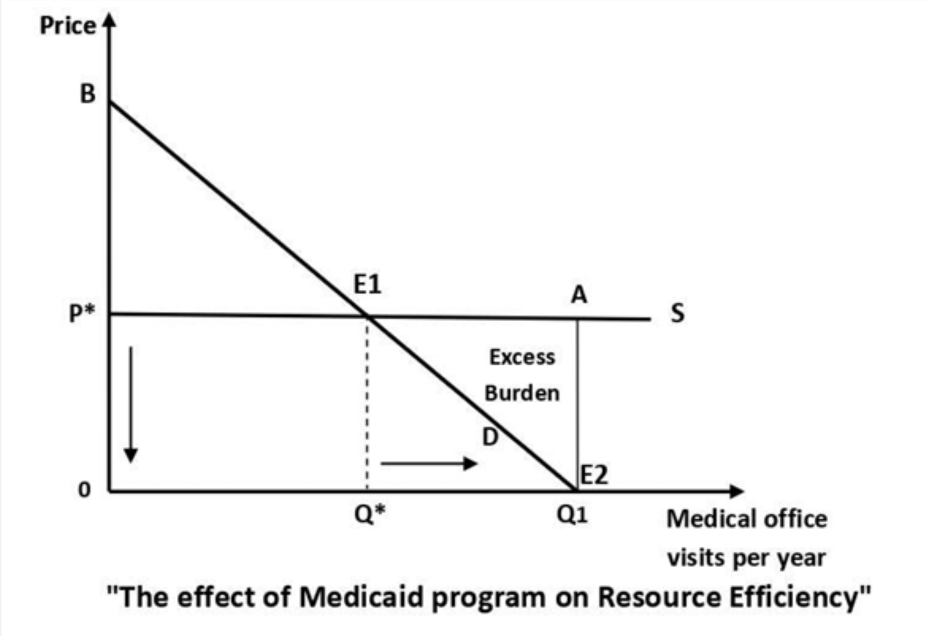
C- Medicaid reduces the effective wage rate in the NR segment

D- Leisure becomes cheaper in the NR segment

Answer: B

📊 GRAPH 9: Effect of Medicaid on Resource Efficiency (Excess Burden)

The Graph:



Key Information:

- Horizontal axis: Quantity of medical services (office visits per year)
- Vertical axis: Price of medical services
- S: perfectly elastic supply curve (competitive industry) at price P*
- D: demand curve for medical services by low-income people
- **Before Medicaid:** equilibrium at E1 (price P*, quantity Q*) — EFFICIENT because marginal benefit = marginal cost
- Consumer surplus before = ΔBE1P*
- **After full Medicaid subsidy:** price to consumer becomes ZERO
- New equilibrium at E2 (price = 0, quantity = Q1 > Q*)
- Consumer surplus after = ΔBE2O
- Gain in consumer surplus = P*E1E2O
- **Overall cost of government subsidy = P*AE2O** (= P* × Q1)
- **Excess burden = ΔE1E2A** = government cost − gain in consumer surplus = P_AE2O − P_E1E2O
- Excess burden measures the loss in efficiency because consumers consume medical services BEYOND the point where marginal benefit equals marginal cost

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شرح الرسمة : الرسمة دي بتوضح الأثر على كفاءة الموارد لما الحكومة تدي دعم كامل (صفر سعر) للرعاية الصحية.
قبل الميديكيد: التوازن عند E1 — سعر P*، كمية Q*. ده السعر الكفء لأن الفائدة الحدية = التكلفة الحدية.
بعد الميديكيد (سعر = صفر للفقراء): الكمية المطلوبة بترتفع لـ Q1 (أكبر من Q*). الناس بتستهلك خدمات طبية حتى لو فايدتها الحدية منخفضة جداً (لأن السعر صفر ماحدش بيحسب تكلفة).
الـ Excess Burden: هو المثلث E1E2A — بيمثل الخسارة في الكفاءة. يعني الحكومة بتدفع أكثر مما المستهلكين بيكسبوا من الفائدة في جزء الاستهلاك الزيادي (من Q* لـ Q1).
ببساطة: الدعم الكامل خلى الناس تاخد أكتر مما تحتاجه فعلاً ← هدر في الموارد.

MCQs:

Q1. Before Medicaid, the equilibrium at E1 (price P, quantity Q) is described as efficient because:**
A- It is the lowest possible price in the market
B- At P*, marginal benefits of medical services equal their marginal costs
C- It maximizes government tax revenue
D- It eliminates all consumer surplus
Answer: B

Q2. After full Medicaid subsidy is introduced, the price of medical services to low-income persons becomes:
A- P* (the market equilibrium price)
B- Half of P*
C- Zero — the government covers the full cost
D- Higher than P* due to increased demand
Answer: C

Q3. The excess burden of the Medicaid subsidy (ΔE1E2A) represents:
A- The total consumer surplus gained from Medicaid
B- The government's profit from providing health services
C- The loss in efficiency because consumers use medical services beyond the efficient quantity Q*
D- The producer surplus lost by health care providers
Answer: C

Q4. The overall cost of the government's Medicaid subsidy equals:

A- P* × Q* only

B- The area P_AE2O = P_ × Q1

C- The triangle ΔBE1P*

D- The area ΔE1E2A only

Answer: B

Q5. The gain in consumer surplus after Medicaid equals:

A- ΔBE1P* (consumer surplus before Medicaid)

B- The area P*E1E2O

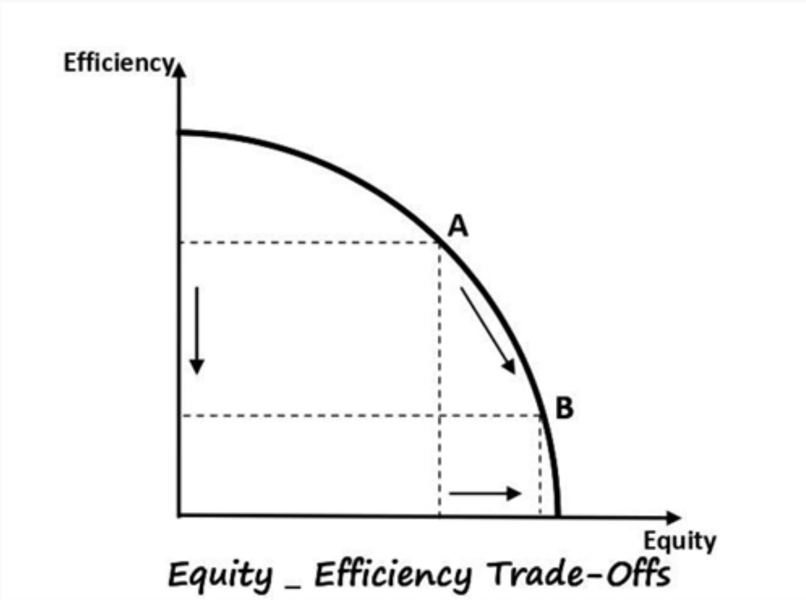
C- The area ΔE1E2A (excess burden)

D- Zero — consumers pay nothing so there is no surplus

Answer: B

📊 GRAPH 10: Equity–Efficiency Trade-Off

The Graph:



Key Information:

- Horizontal axis: Equity (income distribution fairness)
- Vertical axis: Efficiency (productive/allocative efficiency of the economy)

- The curve is concave (bowed outward from origin) — showing the trade-off frontier
- **Point A:** higher efficiency, lower equity
- **Point B:** lower efficiency, higher equity
- Moving from A to B along the curve: a slight increase in equity requires giving up a LOT of efficiency
- Trade-off arises because when government attempts to achieve more equitable income distribution it may:
  - Decrease incentives to work (as seen in unemployment grant and Medicaid)
  - Distort consumption patterns (excess burden)
  - Cause losses in efficiency in market operation
- **General principles to manage trade-offs:**
  - Growth promotion
  - Competitive tendering
  - Control and audit
  - Transparency
  - Public scrutiny

شرح الرسمة : الرسمة دي من أهم الرسومات في الكتاب — بتوضح التعارض الأساسي في السياسة المالية.
المحور الأفقي: العدالة (إيه مدى عدالة توزيع الدخل). المحور الرأسي: الكفاءة (إيه مدى كفاءة عمل الاقتصاد).
المنحنى محدب — بيمثل الحد الأقصى للعلاقة بين الاتنين.
النقطة A: كفاءة عالية، عدالة أقل. النقطة B: عدالة أعلى، كفاءة أقل.
لو الحكومة حبت تنتقل من A لـ B (تزيد العدالة شوية) ← لازم تضحي بكتير من الكفاءة. ليه؟ لأن برامج إعادة التوزيع زي الإعانات والميديكيد بتقلل الحافز على العمل وبتعمل تشوه في الاستهلاك.
الحل مش إنك تلغي البرامج، لكن تديرها صح: شفافية، رقابة، مناقصات تنافسية، وتوجيه الإنفاق نحو النمو.

MCQs:

Q1. In the Equity–Efficiency Trade-Off graph, moving from point A to point B means:
A- Both efficiency and equity increase simultaneously
B- A slight increase in equity requires giving up a large amount of efficiency
C- Efficiency increases while equity decreases
D- The economy moves inside the frontier to a less optimal position
Answer: B

Q2. The Equity–Efficiency trade-off arises in public expenditure programs mainly because:
A- Government programs always increase both equity and efficiency
B- Attempts to achieve more equitable income distribution may decrease work incentives and distort consumption patterns
C- Efficiency and equity are always perfectly complementary goals
D- Private markets naturally solve both equity and efficiency problems
Answer: B

Q3. Which of the following is a real-world example of the Equity–Efficiency trade-off discussed in this chapter?
A- Capital expenditure on roads increasing both GDP and equity
B- Unemployment grants reducing work incentives while trying to help low-income families
C- Investment in education with no side effects
D- Tax cuts that benefit only the wealthy
Answer: B

Q4. Which of the following is NOT listed as a general principle for managing the Equity–Efficiency trade-off?
A- Growth promotion
B- Competitive tendering
C- Eliminating all social welfare programs
D- Transparency and public scrutiny
Answer: C

Q5. The shape of the Equity–Efficiency frontier (concave/bowed outward) indicates:

A- It is easy to gain both equity and efficiency simultaneously

B- The trade-off becomes increasingly costly — gaining more equity requires sacrificing more and more efficiency

C- The relationship between equity and efficiency is linear

> D- Maximum efficiency is achieved only at maximum equity
>
> **Answer: B**